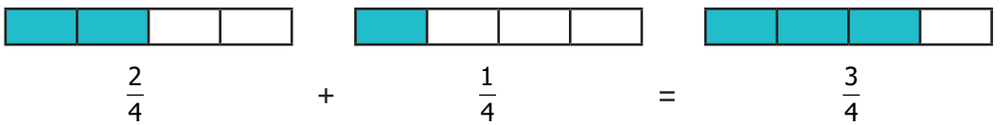
Solution

Ravi's way	Arun's way
Common Multiple of 2 and 4 = 4. Equivalent Fractions of $\frac{1}{2}=\frac{1\times2}{2\times2}=\frac{2}{4}$ Now, add $\frac{1}{2}+\frac{1}{4}=\frac{2}{4}+\frac{1}{4}$ $=\frac{2+1}{4+4}=\frac{3}{8}$ Therefore Venkat has bought $\frac{3}{8}$ litres of milk.	Common Multiple of 2 and 4 = 4. Equivalent Fractions of $\frac{1}{2}=\frac{1\times2}{2\times2}=\frac{2}{4}$ Now, add $\frac{1}{2}+\frac{1}{4}=\frac{2}{4}+\frac{1}{4}$ $=\frac{2+1}{4}=\frac{3}{4}$ Therefore Venkat has bought $\frac{3}{4}$ litres of milk.

The teacher concludes that Arun's way is correct. This can be verified by the following diagram.



In the above illustration note that while adding two like fractions the total number of parts (denominator) remains the same and the two shaded parts (numerator) are added.

Example 6 Add $\frac{2}{3}$ and $\frac{3}{5}$.

Solution These are unlike fractions, aren't they ? So first we need to convert them into like fractions ? Is it possible ? Yes, always. How do we do so ? The common multiple of 3 and 5 is 15. Hence, we find the equivalent fractions of $\frac{2}{3}$ and $\frac{3}{5}$ with denominator 15.

$$\frac{2}{3}=\frac{2\times5}{3\times5}=\frac{10}{15} \qquad \frac{3}{5}=\frac{3\times3}{5\times3}=\frac{9}{15}$$

$$\frac{2}{3}+\frac{3}{5}=\frac{10}{15}+\frac{9}{15}=\frac{19}{15}$$

In the above example, the common denominator is 15 (3×5). Now we observe that the numerator and denominator of the first fraction is multiplied by 5 which is the denominator of the second fraction. In the same way, the second fraction is multiplied by 3 which is the denominator of the first fraction. Now in finding the numerator of both the like fractions, we need to multiply the numerator of the first fraction by 5 and the numerator of the second by 3. In the denominator, 3×5 and 5×3 are of course the same. Thus, the technique of finding the like fraction is called ***Cross Multiplication*** technique.

That is $\frac{2}{3}\times\frac{3}{5}=\frac{(2\times5)+(3\times3)}{3\times5}=\frac{10+9}{15}=\frac{19}{15}$

6th_ Maths _Chapter 1.indd 8 22-10-2019 18:11:22